

De Novo Biosynthesis of Kynurenic Acid Derivative-Based Antagonist Drug Collection in *E. coli*

Yunbin Wu, Li Zhao, Hao Geng, Xianquan Fu, Qinyao Jia, Wenna Gu, Zixi Shang, and Heng Song*

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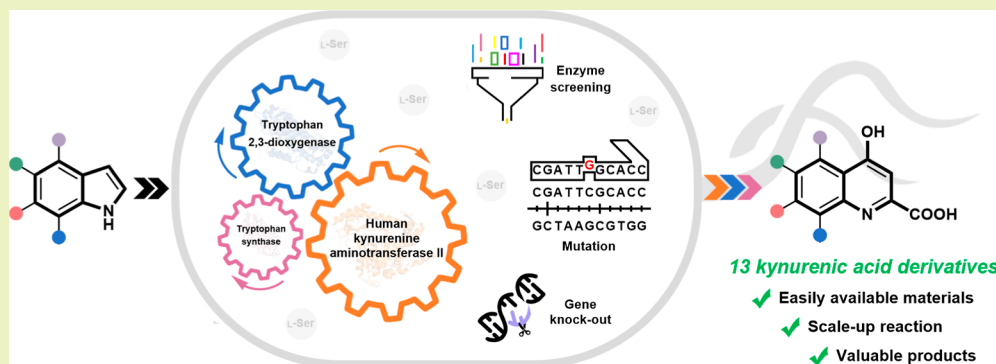
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ABSTRACT: Among heterocycles, quinoline scaffolds are privileged for their numerous biological activities in the pharmaceutical field and have garnered significant attention in the development of novel and potent therapeutic agents. Kynurenic acid (KYNA), also known as 4-hydroxyquinolin-2-carboxylic acid, is a key metabolite of L-tryptophan (L-Trp), which exhibits diverse therapeutic implications in neurodegenerative diseases and certain inflammations. Currently, KYNA and its derivatives are commercially available through chemical synthesis that relies on rigorous reaction conditions and fairly toxic solvents. Therefore, the construction of microbial cell factories offers a promising alternative for the sustainable production of these medicinal moieties. However, no microbial biosynthetic route has been reported so far. In this study, we establish a one-pot, three-enzyme platform in *Escherichia coli* for the diversified biosynthesis of KYNA analogues. As a result, 13 KYNA derivatives with diverse substituent groups (e.g., F, Cl, Br, CH₃, and OCH₃) can be rapidly assembled in a predictable manner with 21–85% isolated yields. And our approach enables translating the reaction parameters from Eppendorf tubes to the 1 L scale, affording 7-Cl-KYNA with 907 mg/L. Moreover, the finding that the enzymatic steps mediated by kynurenine formamidase and kynurenine aminotransferase can be reversed in order undoubtedly provides fresh and exciting cognitive horizons for the L-Trp metabolic pathway. Additionally, targeted engineering of kynurenine aminotransferase (R20G mutant) achieved a 2.5-fold improvement in solubility and a 2.8-fold enhancement in catalytic efficiency, laying a foundation for future structural biological studies, drug-targeting research, and industrial applications. Overall, this is the first report on the microbial production of KYNA and its derivatives and provides an environmentally friendly method for sustainable scale-up synthesis via *E. coli*.

KEYWORDS: kynurenic acid derivatives, antagonist drug, de novo biosynthesis, whole-cell catalysis, quinoline scaffolds

1. INTRODUCTION

Among heterocyclic compounds, quinoline scaffolds are privileged for their numerous therapeutic applications (e.g., antimalarial, antibacterial, antiviral, anticancer, and antituberculosis) in the pharmaceutical field (Figure 1A)^{1–5} and have become an important pharmacophore in contemporary drug discovery.⁶ As representative members of the quinoline family, kynurenic acid (KYNA) and its derivatives are famous for their diverse therapeutic properties against neurodegenerative diseases (e.g., schizophrenia, major depressive disorder, and Alzheimer's disease),^{7–9} especially acting as antagonists of α 7-nicotinic acetylcholine and NMDA receptors to regulate neurotransmission.¹⁰ And studies indicate that KYNA also

plays a role in both gut homeostasis regulation and anti-inflammatory therapy.¹⁰ In fact, the biological significance of KYNA is not limited to these. A recent study shows that leaf beetles can exploit KYNA to detoxify the chemical defenses of poplar trees, highlighting its potential as a novel target for developing selective pest control agents.¹¹ Therefore, devel-

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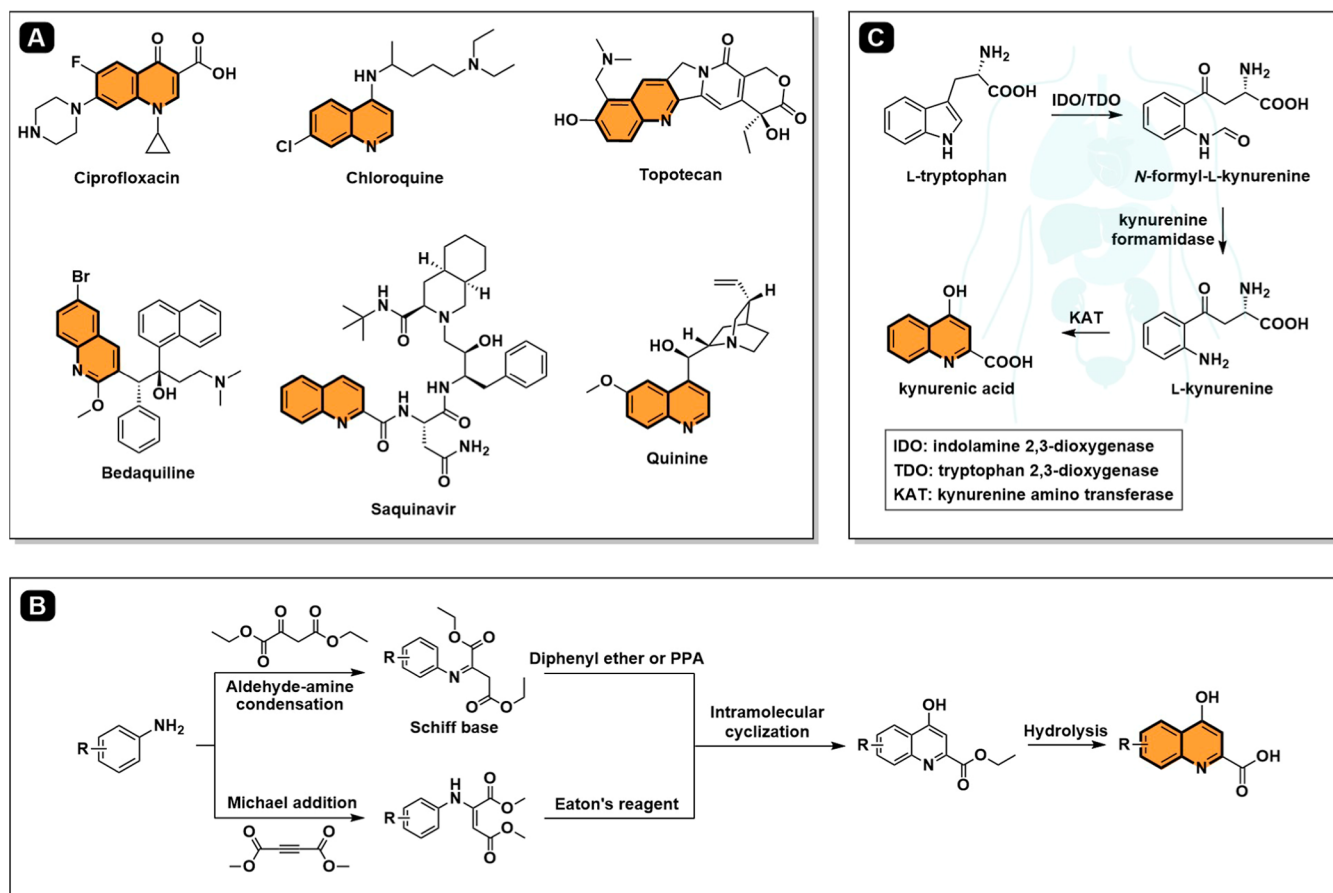


Figure 1. Therapeutic applications of quinoline-based drugs and general synthetic strategies for KYNAs. (A) Market drugs containing quinoline scaffold. (B) Representative synthetic approaches to KYNA scaffolds. (C) The metabolic pathway from L-tryptophan to L-kynurenine in mammals.

oping efficient synthetic methods for KYNA and its derivatives to supply sufficient amounts for in vivo tests is crucial to downstream drug-targeting research and drug lead discoveries.

Currently, KYNA and its derivatives are mainly synthesized via chemical approaches (Figure 1B).^{12–14} Although the chemical community has developed several elegant examples to synthesize this intriguing pharmaceutical core, current methods still possess some limitations such as harsh cyclization conditions, significant side reactions, and fairly toxic solvents. Thus, the large-scale synthesis of KYNA remains highly challenging. To overcome the difficulties in chemical process, whole-cell biocatalysis—in which heterologous biosynthetic pathways are constructed in microbes—offers a promising alternative for the sustainable production of these therapeutic agents.^{15,16} However, no microbial biosynthetic route has been reported so far. This does not mean that the enzymes responsible for the synthesis of KYNA are fully unknown. In mammals, KYNA is a pivotal metabolite of the tryptophan-kynurenine metabolic pathway.¹⁷ And tryptophan (Trp) is converted to KYNA via three enzymatic steps catalyzed by tryptophan 2,3-dioxygenase/indoleamine 2,3-dioxygenase, kynurenine formamidase, and kynurenine aminotransferase (Figure 1C). Hence, the native metabolic processing of Trp to KYNA provides significant inspiration for designing KYNA biosynthetic route. But obtaining structurally diverse KYNAs from Trps is still difficult. This challenge largely stems from the complex handling of various regioselective enzymes and their poor solubility.^{18–22} Directly employing Trp derivatives as substrates is also not feasible due to the costly source materials

and unknown enzyme adaptation.²³ Therefore, redesigning the synthetic route de novo from commercial substrates, coupled with mining enzymes of broad substrate adaptability, offers a highly promising way to advance KYNA biosynthesis.

In this study, we successfully engineered a microbial cell factory for the high-yield production of KYNA and its analogues. Based on the native L-Trp metabolic pathway, we first redesigned an efficient three-enzyme, one-pot cascade for the synthesis of the KYNA scaffold from readily available indole analogues and L-serine by relocating the kynurenine aminotransferase-mediated deamination/cyclization step upstream and introducing trifluoroacetic acid (TFA) for formyl hydrolysis. Then, tailored enzyme mining strategies were used to screen the enzymes for each cascade step. And all three screened enzymes (tryptophan synthase Pf0A9 from *Pyrococcus furiosus*, tryptophan 2,3-dioxygenase TDO from *Xanthomonas campestris*, and kynurenine aminotransferase hKAT II from *Homo sapiens*) exhibited broad substrate promiscuity, affording an array of KYNA derivatives (13 examples with F, Cl, Br, CH₃, and OCH₃ substituents) in 21–85% isolated yields. Finally, through the combination of endogenous substrate degradation pathway disruption, hKAT II solubility engineering, and fusion-protein design, we achieved a high production titer of 907 mg of the target product in a 1 L-scale whole-cell reaction system.

2. MATERIALS AND METHODS

2.1. Reagents, Plasmids, Vectors, and Strains

All chemical reagents were purchased from commercial suppliers (Energy Chemical, Heowns, Innochem, Sigma-Aldrich et al.) if not otherwise stated and were used without further purification. The plasmids (*Pf0A9*, *TDO*, *Tar16*, and *hKAT II*) and vectors (*pCDFDuet-1* and *pRSFDuet-1*) were purchased from GENEWIZ, Inc. The *Escherichia coli* DH5 α and *E. coli* BL21 (DE3) strains were purchased from Sangon Biotech, Inc.

2.2. Analytical Methods

High-performance liquid chromatography (HPLC) analysis was performed for the detection and purification of KYNA and its analogues. ^1H and ^{13}C NMR spectra were recorded on a Bruker ADVANCE III 600 MHz spectrometer. High-resolution mass spectra were obtained from a Thermo Scientific Q Exactive mass spectrometer in an electrospray ionization (ESI) mode.

2.3. KYNA Analogues Produced by *Pf0A9/TDO/hKAT II/TFA*-Catalyzed Cascade Reactions (In Vitro)

The indole analogues (0.5 mM), L-serine (0.55 mM), PLP (0.1 mM), sodium ascorbate (0.1 mM), hemin (0.05 mM), 2-ketoglutaric acid (0.05 mM), *Pf0A9* (0.005 mM), *TDO* (0.005 mM), and *hKAT-II* (0.005 mM) were added to a 50 mL centrifuge tube, followed by enough KPi buffer (100 mM, pH = 8.0) to make the final volume 30 mL. The tube was opened and placed in a metal bath that had been equilibrated to 30 °C for 6 h. After that, TFA (1% (v/v) = 300 μL) was added, and the reaction mixture was kept at 30 °C for 1 h. Then the reaction mixture was quenched by the addition of an equal volume of HPLC grade methanol. And the resultant was evaporated under reduced pressure (Note: the purpose of this step is to remove methanol to reduce the total volume of HPLC injection), centrifuged at 12,000 rpm for 10 min, and filtrated until further analysis and purification by HPLC. In this part, we have attempted various indole analogues, including 4-F-indole, 5-F-indole, 5-Cl-indole, 5-Br-indole, 5-CH $_3$ -indole, 5-OCH $_3$ -indole, 6-F-indole, 6-Cl-indole, 6-Br-indole, 6-CH $_3$ -indole, 6-OCH $_3$ -indole, 5,6-difluoro-indole, and 7-F-indole. All products were characterized by ^1H NMR. Due to the poor solubility of some products (e.g., 5-F-KYNA, 7-CH $_3$ -KYNA, and 8-F-KYNA), which made ^{13}C NMR characterization challenging under current experimental conditions, high-resolution characterization was performed instead. When performing ^{13}C NMR measurements, a small amount of NaOH can be added to increase the solubility of the product (Note: do not add too much NaOH, as it may cause shimming failure) (Figures S6–S40 in the Supporting Information).

2.4. Plasmid Construction

Gene fragments and vectors were amplified with the corresponding primers listed in Tables S10–S19 in the Supporting Information. PCR products were identified by gel electrophoresis and purified with a gel extraction kit. After that, the Gibson assembly was applied to connect the gene fragments to the corresponding vector. The products were transformed into *E. coli* DH5 α . The construction results were confirmed by Sanger sequencing.

2.5. Construction of the *TnaA* Tryptophanase Knockout *E. coli* BL21(DE3) Strain

The *TnaA* tryptophanase knockout strain of *E. coli* BL21(DE3), named *E. coli* BL21 $\Delta tnaA$ (DE3), was constructed for the production of KYNA and its derivatives. The *E. coli* BL21 $\Delta tnaA$ (DE3) strain was constructed using a previously developed two-plasmid-based CRISPR-Cas9 system.²⁴ The two key plasmids, pEcCas and pEcgRNA, were purchased from GENEWIZ, Inc. The pEcgRNA-*tnaA*-N20 plasmid was constructed by replacing the *ccdB* gene (pEcgRNA plasmid) with the *tnaA*-N20 sequence (<https://chopchop.cbu.uib.no/>) using the corresponding primers (Table S6 in the Supporting Information). The resulting pEcgRNA-*tnaA*-N20 plasmid was sequenced to confirm the introduction of the *tnaA*-N20 sequence. Then the *tnaA*-Up + *tnaA*-Down fragment was amplified from the *E.*

coli genome using the corresponding primers (Table S6 in the Supporting Information). Finally, the *E. coli* BL21 $\Delta tnaA$ (DE3) strain was constructed by using the pEcCas plasmid, pEcgRNA-*tnaA*-N20 plasmid, and *tnaA*-Up + *tnaA*-Down fragment according to a previously developed procedure.²⁴

2.6. Whole-Cell System for the Production of 7-Cl-KYNA (1 mL/1 L-Scale)

E. coli BL21 $\Delta tnaA$ (DE3) cells transformed with the expression plasmid constructed above were inoculated into ~ 5 mL of LB medium containing 50 $\mu\text{g}/\text{mL}$ antibiotic (kanamycin for the pRSFDuet-1 series; streptomycin for the pCDFDuet-1 series) and grown 6 h at 37 °C and 220 rpm. Then, 1 L of LB medium containing 50 $\mu\text{g}/\text{mL}$ antibiotic was inoculated with ~ 5 mL of 6 h culture. The culture was incubated at 37 °C, 220 rpm until the OD $_{600}$ reached ~ 0.8 , at which point δ -aminolevulinic acid and IPTG stock solutions were added to final concentrations of 1 and 0.5 mM, respectively, to induce protein overexpression at 16 °C, 220 rpm for 16 h. Cells were harvested by centrifugation at 4 °C, 3500 rpm for 15 min. Then, the cells were resuspended in 100 mM KPi buffer (pH = 8.0) and diluted down to OD $_{600}$ = 50. The whole-cell catalytic reaction was performed at a final volume of 1.0 mL in a 10 mL tube or 1 L in a 2.5 L flask.

2.6.1. For 1 mL Reaction System. 6-Cl-indole (5.0 mM; dissolved in 5% dimethyl sulfoxide (DMSO) (v/v; 50 μL); added in two batches at 2 h intervals), L-serine (5.5 mM; dissolved in KPi buffer; added all at once), PLP (1.0 mM; dissolved in KPi buffer; added all at once), sodium ascorbate (1.0 mM; dissolved in KPi buffer; added all at once), hemin (0.5 mM; dissolved in $\sim 0.1\%$ DMSO (v/v; 1 μL); added all at once), 2-ketoglutaric acid (0.5 mM; dissolved in $\sim 0.1\%$ DMSO (v/v; 1 μL); added all at once), and the above bacterial suspension (OD $_{600}$ = 50; added 950 μL) were added to a 10 mL tube. Note: the total added volume of L-serine, PLP, sodium ascorbate, hemin, and 2-ketoglutaric acid should not exceed 5 μL . Thus, the stock solution should be prepared at high concentration to minimize the addition volume.

2.6.2. For 1 L Reaction System. 6-Cl-indole (5.0 mM containing 0.5 mM 2-ketoglutaric acid; dissolved in 5% DMSO (v/v; 50 mL); added in two batches at 2 h intervals), L-serine (5.5 mM; added the compound powder directly; added all at once), PLP (1.0 mM; added the compound powder directly; added all at once), sodium ascorbate (1.0 mM; added the compound powder directly; added all at once), hemin (0.5 mM; dissolved in $\sim 0.1\%$ DMSO (v/v; 1 mL); added all at once), and the above bacterial suspension (OD $_{600}$ = 50; added 950 mL) were added to a 2.5 L flask.

The tube/flask was opened and placed in a metal bath/shaker that had been equilibrated to 30 °C for 16 h. After that, TFA (1% (v/v) = 10 μL for 1 mL reaction system; 0.5% (v/v) = 5 mL for 1 L reaction system) was added, and the reaction was kept at 30 °C for 1 h (for 1 mL reaction system)/3 h (for 1 L reaction system). After the reaction finished, the reaction mixture was centrifuged at room temperature (12,000 rpm for 10 min).

2.6.3. For 1 mL Reaction System. The centrifugal supernatant was mixed with an equal volume of HPLC grade methanol, filtrated, and analyzed by HPLC.

2.6.4. For 1 L Reaction System. The centrifugal supernatant was lyophilized to ~ 25 mL, then mixed with 25 mL of HPLC grade methanol, filtered, and purified by HPLC.

3. RESULTS AND DISCUSSION

3.1. A De Novo Designed Three-Enzyme, One-Pot Cascade for Synthesizing KYNA Scaffold

Although the native metabolic processing of L-Trp into KYNA in mammals provides significant inspiration for designing KYNA biosynthetic route, starting from readily available L-Trp still possesses practical challenges such as complicated operations caused by the use of distinct regioselective halogenases for product diversification, which, in turn, limits the substituent scope chiefly to chloro and bromo groups.^{18–22}

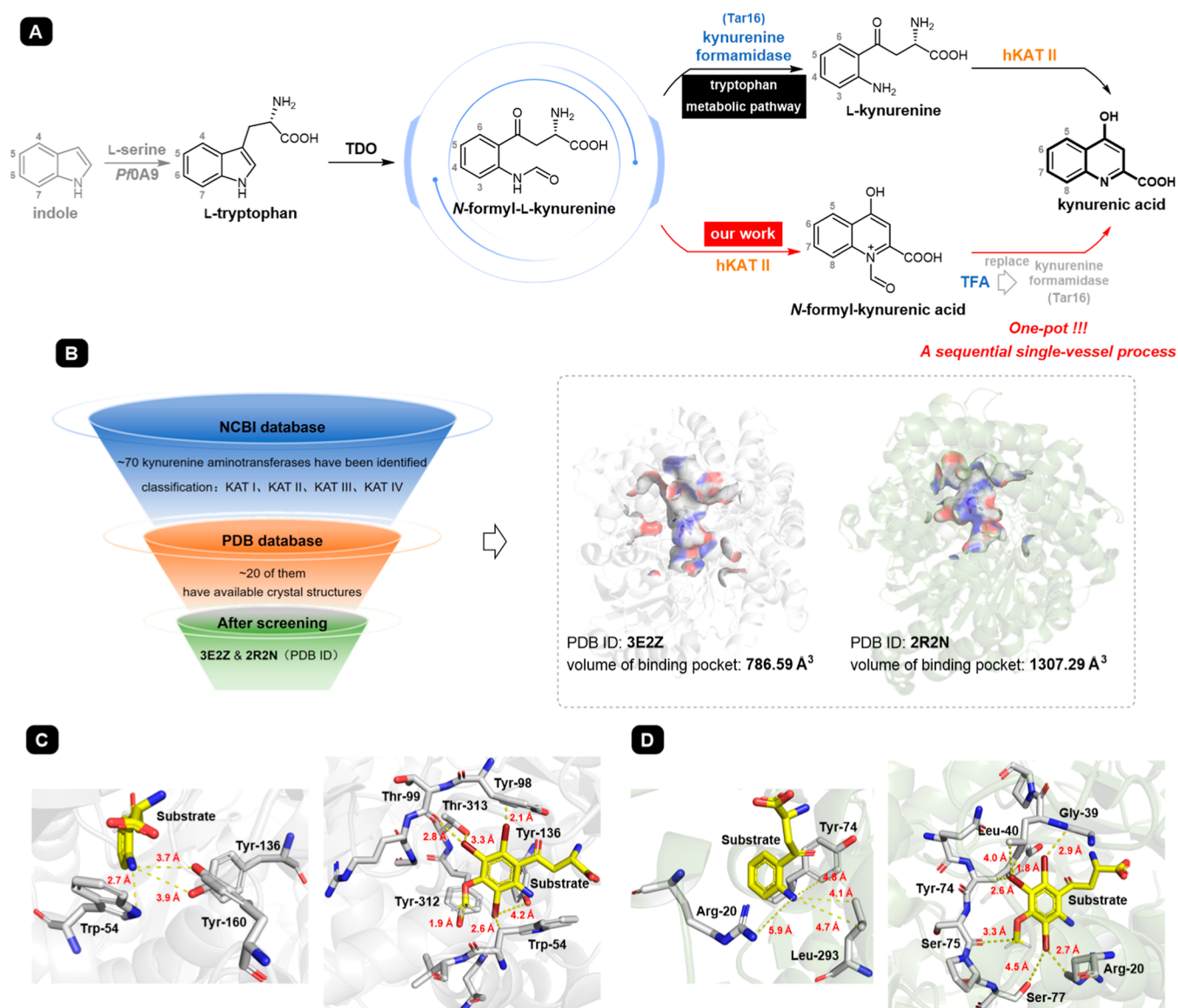


Figure 2. A de novo designed three-enzyme, one-pot cascade for synthesizing KYNA scaffold. (A) Synthetic approaches toward KYNA skeleton. By replacing kynurenine formamidase with TFA and reversing the catalytic order of hKAT II and TFA, the entire synthetic route was simplified to a sequential single-vessel, three-enzyme cascade process. (B) Comparison of the catalytic pocket sizes of mouse kynurenine aminotransferase (PDB ID: 3E2Z) and human kynurenine aminotransferase (PDB ID: 2R2N). Among more than 70 known kynurenine aminotransferases, only 3E2Z and 2R2N have substrate-bound cocrystal structures. (C) Stereoview of amino acid residues within 5 Å of the substrate amino group (left) and benzene ring (right) in mouse kynurenine aminotransferase. (D) Stereoview of amino acid residues within 5 Å of the substrate amino group (left) and benzene ring (right) in human kynurenine aminotransferase. In (C,D), the substrate is colored yellow. The critical residues around amino group and benzene ring are labeled in gray and shown by sticks. According to the results calculated by Jerkwin tool in Table S2, we replaced the H atoms on the benzene ring of the substrate with sterically bulkier groups (e.g., 3-Br, 4-OCH₃, 5-Br, and 6-Br) and measured their distances to the surrounding residues to precisely assess the volume of the binding pocket.

Of particular note is the preparation of KYNA analogues directly from corresponding L-Trp precursors, in which the prohibitively expensive L-Trp derivatives will be the biggest obstacle for the next large-scale production. To develop an efficient and economical operation process for the KYNA skeleton, we leverage the L-Trp metabolic pathway, using readily available indole analogues and L-serine as original sources to furnish the requisite tryptophan intermediates. Our design derives the substituted components of KYNA from indole substrates, thereby effectively circumventing the reliance on diverse halogenases and costly L-Trp derivatives (Figure 2A). Notably, the highly engineered tryptophan synthase

(Pf0A9, from *P. furiosus*)²⁵ and tryptophan 2,3-dioxygenase (TDO, from *X. campestris*)²³ exhibit a relatively broader substrate spectrum (≥18 examples) to afford substituted N-formyl-L-kynurenes in moderate to excellent isolated yields, laying a foundation for the diverse synthesis of KYNA and its analogues. Given that the substrate scope of kynurenine formamidase and kynurenine aminotransferase remains largely unexplored, we next aim to further reduce the number of enzymes employed in the cascade to minimize downstream engineering efforts. The formyl hydrolysis step can also be achieved by adding a catalytic amount of (e.g., ≤1% (v/v)) TFA,²⁶ which gives the target products in nearly quantitative

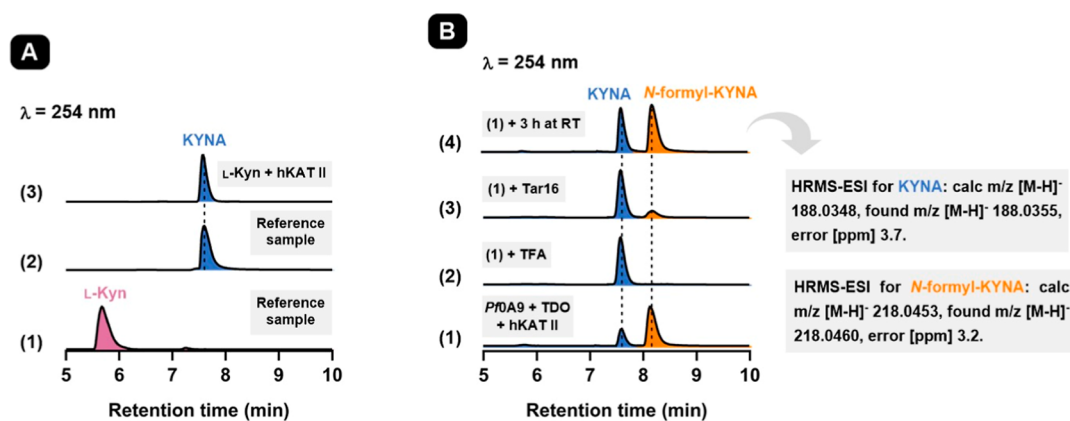


Figure 3. Exploration of the synthetic route to KYNA. (A) Catalytic activity test of hKAT II toward L-kynurenine. (B) Investigation of hKAT II substrate tolerance toward N-formyl-KYNA and verification of the feasibility of reversing the hKAT II and Tar16 reaction sequence.

yield regardless of the substrate substituents. Additionally, TFA also acted as a workup agent to precipitate both proteins and bacterial cells. Therefore, a chemical transformation was ultimately employed for the formyl hydrolysis step, super-separating the initial biological strategy. However, introducing TFA after the TDO enzymatic step thwarts the one-pot cascade plan as it severely hampers the catalytic activity of the subsequently added kynurenine aminotransferase. To solve this problem, we prioritized the enzymatic step catalyzed by hKAT II over the TFA-mediated hydrolysis process and spatiotemporally separated these two procedures. Specifically, PfoA9, TDO, and hKAT II were added first to generate N-formyl-KYNA, then followed by enough TFA to yield the final target product KYNA, thus enabling the entire synthetic pathway to be performed as a sequential single-vessel process (Figure 2A).

With an efficient cascade in hand, we next focus on identifying a kynurenine aminotransferase to serve as a general biocatalytic platform for the synthesis of a range of KYNA analogues. Notably, compared with L-kynurenine, the formyl group imparts greater steric bulk to N-formyl-L-kynurenine, which implies that the kynurenine aminotransferase should possess a relatively large active cavity. Moreover, it is beneficial if the crystal structure of the kynurenine aminotransferase has been fully illustrated, thus laying a foundation for subsequent active-site mutagenesis. Natural evolution has provided us with approximately 70 types of kynurenine aminotransferases. These enzymes can be subdivided into four classes (kynurenine aminotransferase I, II, III, and IV), and all of them can translate L-kynurenine to the corresponding KYNA (Figure 2B).²⁷ However, only about 20 of them have crystal structures. Considering that the results obtained from molecular docking software have inherent errors and to construct a high-quality mutant library to reduce the subsequent screening workload, kynurenine aminotransferase from *H. sapiens* named hKAT II (PDB ID: 2R2N)²⁸ and kynurenine aminotransferase from *Mus musculus* named mKAT III (PDB ID: 3E2Z)²⁹ were chosen for further investigation due to their substrate-bound cocrystal structures (Table S1 in the Supporting Information). Aided by the Jernwin tool, we calculated the molecular sizes of N-formyl-L-kynurenine substituted with different groups (e.g., F, Cl, Br, CH₃, and OCH₃) at various positions (e.g., 3-, 4-, 5-, and 6-positions) on the benzene ring (Table S2 in the Supporting Information). The calculation results indicated that the active cavity volume of the protein should preferably exceed 650 Å³

in order to achieve higher reactivity toward sterically hindered groups (Table S2 in the Supporting Information). With the support of the Proteins Plus tool, we found that the binding pocket sizes of hKAT II (1307.29 Å³) and mKAT III (786.59 Å³) are both above this threshold (Figure 2B). But after measuring the distances from the substrate amino group and benzene ring to the surrounding residues with PyMol software, hKAT II was ultimately chosen for subsequent experiments due to its comparatively spacious reactive space (Figure 2C,D).

To validate our designed biosynthetic route, PfoA9, TDO, Tar16 (a kynurenine formamidase from the marine bacterium *Saccharomonospora* sp. CNQ-490),³⁰ and hKAT II enzymes were individually overexpressed in *E. coli* and purified by Ni-NTA affinity chromatography (Figure S1 in the Supporting Information). In previous work, we have fully investigated the catalytic activities and substrate scopes of PfoA9 and TDO.^{23,26} Therefore, in this study, we will focus on characterizing the catalytic properties of Tar16 and hKAT II. We measured the catalytic activity of hKAT II toward L-kynurenine in a single-step enzymatic reaction (Figure 3A). As expected, hKAT II exhibited excellent catalytic performance (~100% conversion determined by HPLC), which was further confirmed by NMR spectroscopy (Figures S2 and S3 in the Supporting Information). Inspired by this result, we next investigated whether hKAT II could accept the more sterically hindered N-formyl-L-kynurenine as a substrate. However, the instability of N-formyl-L-kynurenine, which was partially deformylated to L-kynurenine at room temperature,³⁰ necessitated an in situ generation and consumption strategy for its reactivity study. Encouragingly, the PfoA9/TDO/hKAT II cascade system converted indole and L-serine to the corresponding N-formyl-KYNA in near-quantitative yield, along with only a trace of the deformylated KYNA product (Figures 3B and S4 and S5 in the Supporting Information). And the subsequent TFA treatment resulted in a complete deformylation of N-formyl-KYNA to KYNA (Figure 3B). The above experimental results indicated that hKAT II showed good tolerance to N-formyl-L-kynurenine, which aligned with our previous calculation data (Figure 2B and Table S2 in the Supporting Information) that the molecular size of N-formyl-L-kynurenine (520.78 Å³) was far smaller than the active cavity volume of hKAT II (1307.29 Å³). Although TFA proved to be efficient at deformylating N-formyl-KYNA, we also wondered whether Tar16 could achieve the same result. In the native L-Trp metabolic pathway, the substrate of Tar16 is N-formyl-L-

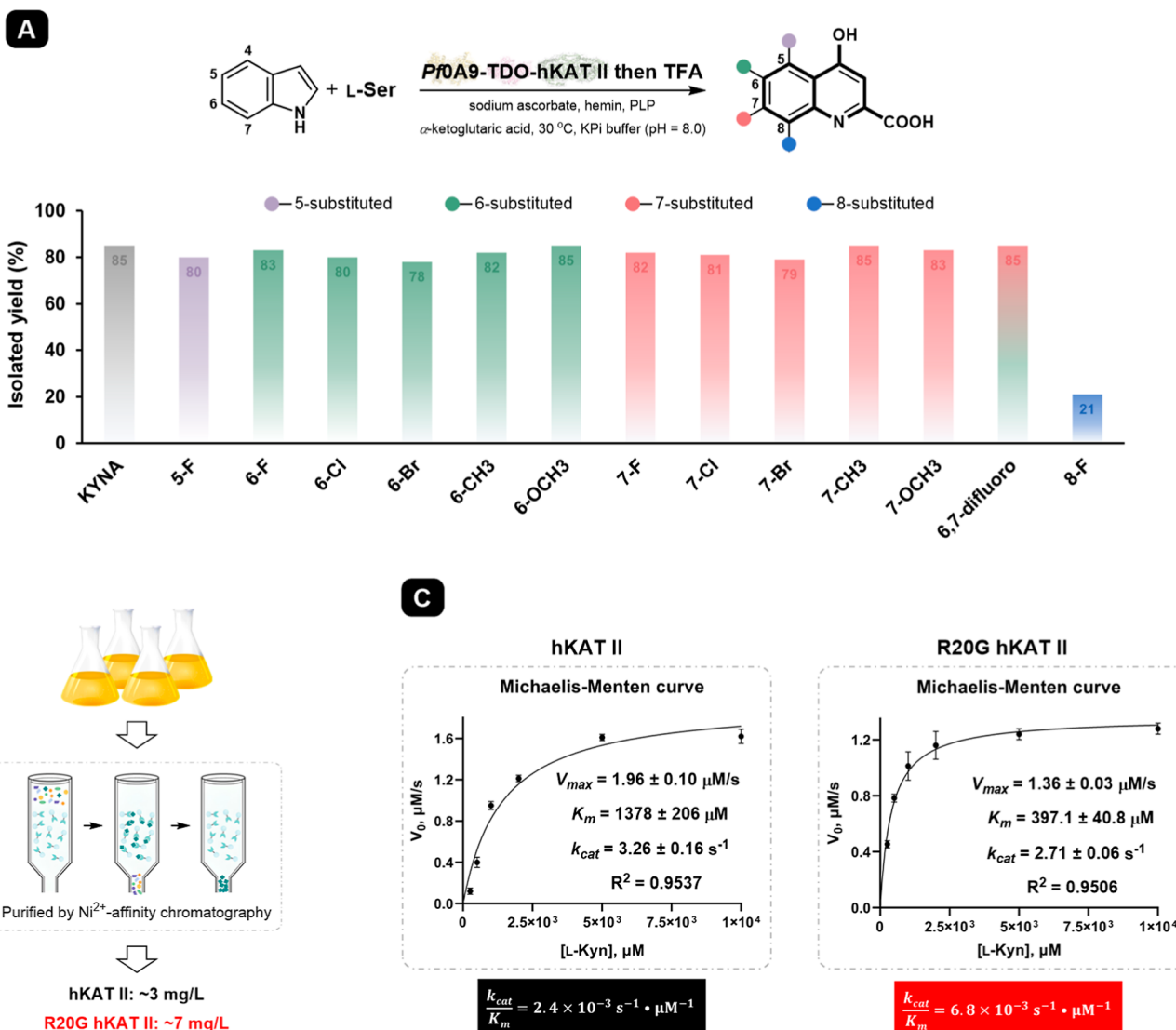


Figure 4. Diversity-oriented biosynthesis of KYNA and its derivatives. (A) KYNA analogues produced by Pf0A9/TDO/hKAT II/TFA-catalyzed cascade reactions. All KYNA analogues were isolated in >75% yield, except for 8-F-KYNA. (B) The soluble expression levels of hKAT II (wild type) and R20G hKAT II (mutant). The R20G mutant resulted in a 2.5-fold increase in protein solubility. (C) The Michaelis–Menten curves were used to determine kinetic parameters of hKAT II and R20G hKAT II. The R20G mutant led to a 2.8-fold improvement in catalytic efficiency.

kynurenine, and the substrate tolerance of Tar16 toward *N*-formyl-KYNA has never been explored. Based on the Pf0A9/TDO/hKAT II cascade system, we added Tar16 after *N*-formyl-KYNA formed. When incubation for 3 h at 30 °C, Tar16 could also deformylate *N*-formyl-KYNA to KYNA (>90% conversion determined by HPLC), albeit with a slightly lower conversion compared to TFA (Figure 3B). Considering the instability of *N*-formyl-KYNA, a control experiment without TFA/Tar16 was conducted (Figure 3B). Approximately 50% spontaneous deformylation to KYNA was observed in the control under identical conditions, confirming that TFA/Tar16 did promote the deformylation process. Therefore, Tar16 can use both *N*-formyl-L-kynurenine and *N*-formyl-KYNA as substrates, which implies that the enzymatic steps mediated by Tar16 and hKAT II in the native *L*-Trp metabolic pathway can be reversed in order. And the interchangeability between Tar16 and hKAT II reveals unprecedented flexibility in the *L*-Trp metabolic pathway and may offer new insight into its mechanistic exploration. However, the instability of *N*-formyl-L-kynurenine precluded

its use as a substrate for the kinetic characterization of hKAT II and Tar16.

3.2. Diversity-Oriented Biosynthesis of KYNA and Its Derivatives

Previous studies of hKAT II have mainly focused on the metabolic mechanisms; therefore, its catalytic function remains only partially elucidated. Considering that the 6- and 7-positions of KYNA are typically targeted sites for structural modification in organic chemistry,^{31,32} we first attempted to introduce halogen substituents (e.g., Cl and Br) there to facilitate further derivatization (Figure 4A). Surprisingly, 5-Cl/5-Br/6-Cl/6-Br-indole all participated smoothly in this enzymatic cascade, providing the corresponding products in excellent isolated yields (78–81%). Besides, the electron-rich groups (e.g., CH₃ and OCH₃) were also examined and afforded the corresponding 6-CH₃-, 6-OCH₃-, 7-CH₃-, and 7-OCH₃-KYNA in 82–85% isolated yields (Figure 4A). All these results demonstrate that the steric hindrance and electronic effects at the 6- and 7-positions are negligible due

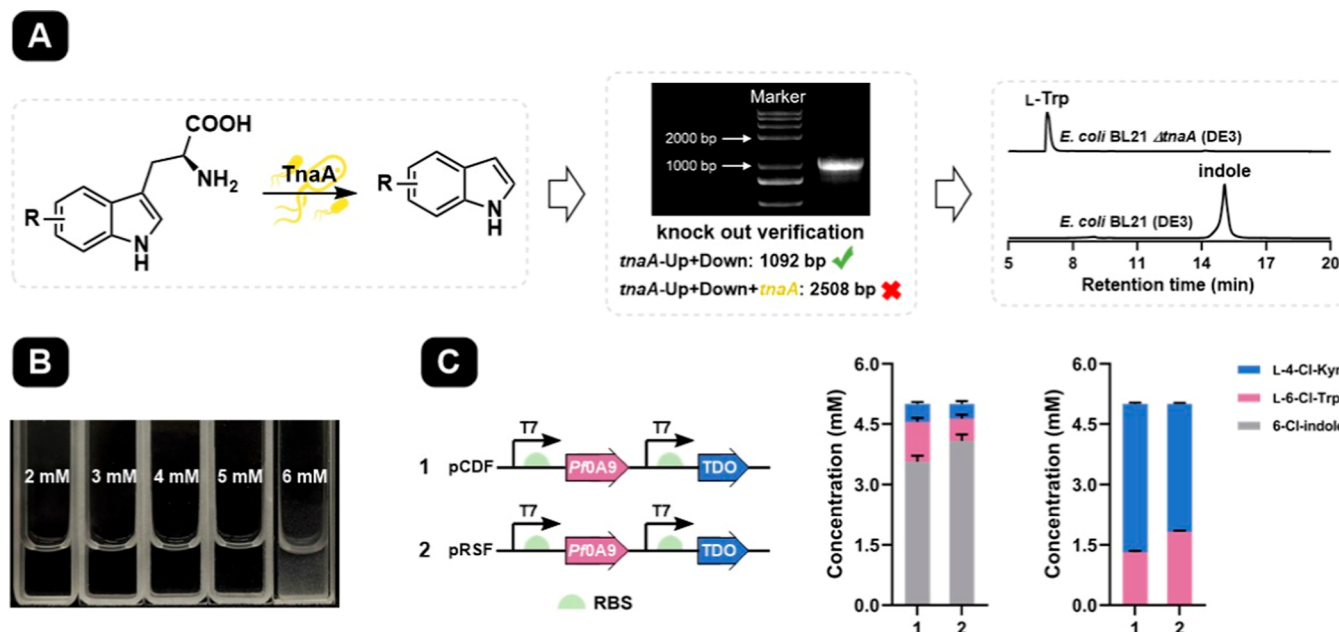


Figure 5. Large-scale experiment preparation. (A) Construction of the TnaA tryptophanase knockout *E. coli* BL21(DE3) strain. (B) The solubility limit of the 6-Cl-indole substrate in 5% DMSO is 5 mM. (C) Construction of *E. coli* cells coexpressing Pf0A9 and TDO for L-4-Cl-Kyn production. pCDFDuet-1 and pRSFDuet-1 vectors were used for dual-gene coexpression.

to the excellent tolerance for diverse substituents (e.g., Cl, Br, CH₃, and OCH₃). Recently, organofluorine molecules have attracted extensive attention owing to their high lipophilicity, metabolic stability, and unique electronic properties.³³ And introducing fluorine atoms or fluorine-containing moieties into bioactive molecules constitutes a fruitful synthetic strategy to improve their pharmacological characteristics in modern drug discovery. Accordingly, we assessed the substrate tolerance of hKAT II toward a series of fluorinated *N*-formyl-L-kynurenine analogues (e.g., 3-F, 4-F, 5-F, 4,5-difluoro, and 6-F). Encouragingly, most substituents, such as 4-F, 5-F, 4,5-difluoro, and 6-F were well tolerated, yielding the corresponding 7-F-, 6-F-, 6,7-difluoro-, and 5-F-KYNA derivatives in 80–85% isolated yields (Figure 4A). In contrast, only 3-F performed poorly, affording 8-F-KYNA in a merely 21% isolated yield (Figure 4A). This unsatisfactory result is attributed to hKAT II because neither 7-F-indole nor L-7-F-Trp was detected by HPLC. Considering that the function of the hKAT II is deamination/cyclization, the F atom at position 3 of the *N*-formyl-L-kynurenine (corresponding 8-position of the final product) may form a hydrogen bonding interaction with the adjacent –NHCHO, thereby affecting the nucleophilicity of the N atom during cyclization and resulting in poor reaction performance. Because wild-type TDO showed no reactivity toward L-4-Cl/Br/CH₃/OCH₃-Trp (corresponding 5-position of the final product) and L-7-Cl/Br/CH₃/OCH₃-Trp (corresponding 8-position of the final product), we were unable to obtain corresponding *N*-formyl-L-kynurenine analogues bearing these substituents and thus could not explore the reactivity of the downstream hKAT II. All KYNA analogues were confirmed by NMR spectroscopy, as shown in Figures S6–S40 in the Supporting Information.

To further improve the yield of 8-F-KYNA, we next conducted structure-guided protein engineering studies on hKAT II. With a substrate-bound cocrystal structure (PDB ID: 2R2N) in hand, we employed a rational site-directed mutagenesis strategy to construct its variant libraries. Based

on the residues marked by PyMol software around L-kynurenine within 5 Å (Figure S41 in the Supporting Information), only Arg20 is close to the 3-position (corresponding 8-position of the final product) of the substrate. Alanine scanning was performed first to find hotspots for further protein engineering. Regrettably, the R20A mutant failed to enhance the yield of 8-F-KYNA (~23% yield as determined by HPLC). And the further attempt to introduce an even smaller side chain by mutating Arg20 to Gly (R20G) was also unsuccessful (~25% yield determined by HPLC). We also tried to mutate Arg20 to other residues (e.g., Asn and Gln) that are structurally similar to Arg but have shorter side chains. Unfortunately, no positive results were observed. Other bulky residues, such as Phe, Trp, and His, were excluded because their side chains would cause an obvious steric clash with the fluorine atom. Although no mutation could further improve the yield of 8-F-KYNA, the R20G variant conferred a 2.5-fold increase in protein solubility (Figures 4B and S42 in the Supporting Information). In *E. coli*, the expression of human kynurenine aminotransferase was challenging because the enzyme typically forms inclusion bodies,²⁸ limiting the soluble protein yield to ~3 mg/L. Although the total expression level of the R20G mutant decreased compared with the wild-type hKAT II, the amount of soluble protein in the supernatant increased (up to ~7 mg/L). Therefore, our finding will undoubtedly advance future structural biological studies and drug-targeting research on hKAT II. In addition, we also investigated the substrate scope of R20G hKAT II toward a range of substituted L-Kyn analogues and found that the variant exhibited comparable HPLC yields to the wild-type enzyme (Figure S43 in the Supporting Information). Finally, kinetic studies were performed to elucidate the catalytic properties of WT hKAT II and the R20G mutant by monitoring the formation of the product using HPLC at 254 nm (Figure 4C and Tables S4 and S5 in the Supporting Information). The kinetic parameters K_m and k_{cat} for WT hKAT II toward L-Kyn were $1378 \pm 206 \mu\text{M}$

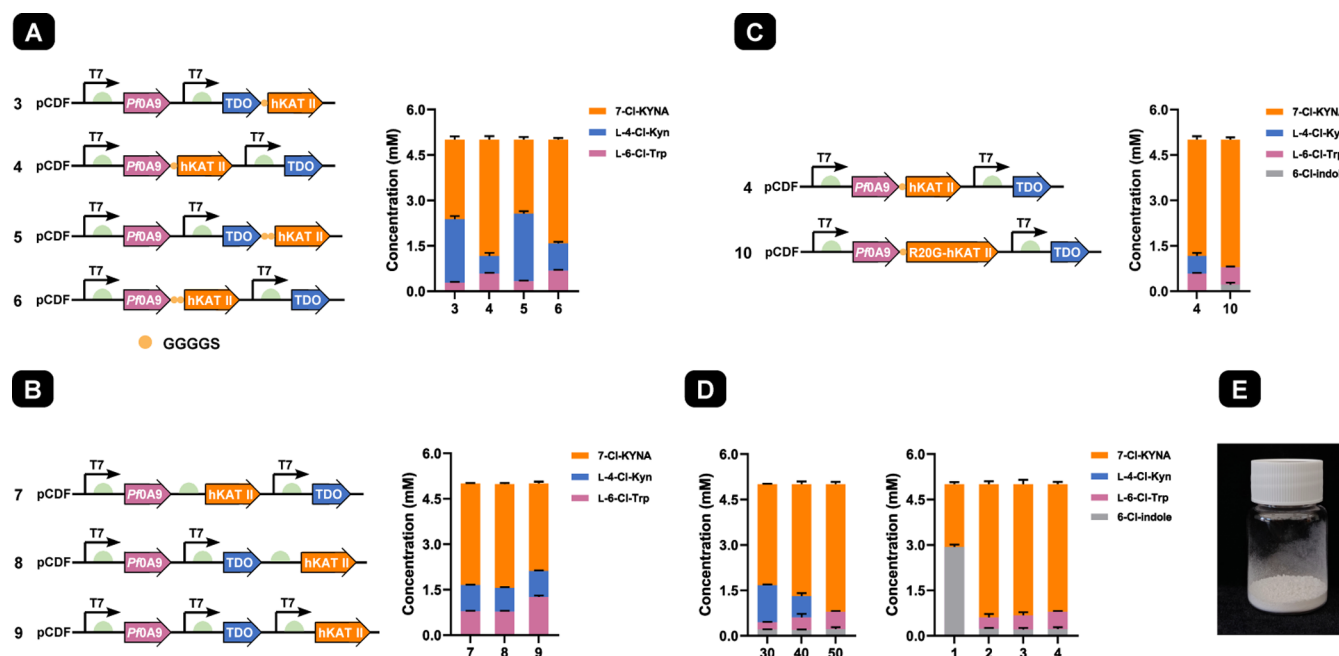


Figure 6. Large-scale synthesis of 7-Cl-KYNA in an efficient three-enzyme, one-pot manner. (A) Construction of *E. coli* cells coexpressing *Pf0A9*, TDO, and hKAT II for 7-Cl-KYNA production. G_4S and $(G_4S)_2$ were used to fuse *Pf0A9* and TDO to the N-terminus of hKAT II to enhance its solubility. (B) Influence of ribosome binding site (RBS) engineering and T7 promoter copy number on 7-Cl-KYNA production. (C) Comparing the catalytic efficiency of hKAT II and R20G hKAT II variant on 7-Cl-KYNA production. (D) Effects of cell density and 6-Cl-indole feeding frequency on 7-Cl-KYNA production. (E) 907 mg of 7-Cl-KYNA from a 1 L-scale reaction system.

and $3.26 \pm 0.16 \text{ s}^{-1}$, respectively ($k_{\text{cat,app}}/K_{\text{m,app}} \sim 2.4 \times 10^{-3} \text{ s}^{-1} \cdot \mu\text{M}^{-1}$). The K_{m} and k_{cat} values for R20G hKAT II toward L-Kyn were $397.1 \pm 40.8 \mu\text{M}$ and $2.71 \pm 0.06 \text{ s}^{-1}$, respectively ($k_{\text{cat,app}}/K_{\text{m,app}} \sim 6.8 \times 10^{-3} \text{ s}^{-1} \cdot \mu\text{M}^{-1}$). Thus, The R20G mutant led to a 2.8-fold enhancement in catalytic efficiency, laying a foundation for future industrial applications.

3.3. Large-Scale Synthesis of 7-Cl-KYNA in an Efficient Three-Enzyme, One-Pot Manner

To demonstrate the applicability of our approach in practical manufacturing, we applied our protocol to synthesize 7-Cl-KYNA on a large scale in *E. coli*. Tryptophanase, encoded by the endogenous *tnaA* gene in *E. coli*, catalyzes the degradation of L-Trp analogues to the corresponding indoles, ammonia, and pyruvate.³⁴ Since L-Trp analogues serve as important precursors for synthesizing KYNA derivatives in our cascade, we knocked out the *tnaA* gene using CRISPR-Cas9 gene-editing technology to block this competing degradation pathway (Figures S4 and S44–S48 in the Supporting Information, and Tables S6–S9 in the Supporting Information). As expected, the *E. coli* strain with the TnaA tryptophanase knocked out no longer consumed Trp and its derivatives (Figures S4 and S48 in the Supporting Information). With the *E. coli* BL21 ΔtnaA (DE3) strain established, we first attempted to coexpress *Pf0A9* and TDO in this host. As shown in Figure S5C, *Pf0A9* and TDO were constructed on plasmid vectors with different copy numbers (Figures S53–S56 in the Supporting Information, Tables S10 and S11 in the Supporting Information), including pCDFDuet-1 (~20 copies) and pRSFDuet-1 (~100 copies).³⁵ Considering that DMSO, used to dissolve organic indole substrates, could affect the growth and proliferation of *E. coli* at high concentrations (Figure S49 in the Supporting Information),³⁶ the DMSO content in the whole-cell reaction was fixed at 5% (v/v), at which up to 5 mM of 6-Cl-indole can be dissolved

(Figure S5B). Accordingly, the whole-cell biocatalysis was performed in 1.0 mL of KPi buffer (100 mM, pH = 8.0) containing 5 mM substrate at 30 °C with the cell density adjusted to $\text{OD}_{600} = 50$. Unexpectedly, although pCDFDuet-1-*Pf0A9*-TDO yielded 1.25-fold more product than pRSFDuet-1-*Pf0A9*-TDO, the titer of L-4-Cl-kynurenine (L-4-Cl-Kyn) was merely 0.45 mM, with a substantial substrate remaining (Figure S5C). However, SDS-PAGE analysis showed that the two coexpressed proteins were largely soluble (Figures S54 and S56 in the Supporting Information), rather than forming inclusion bodies, indicating that the low substrate conversion could not be attributed to insufficient soluble proteins. A comprehensive literature search revealed that indole scaffolds possess moderate to potent activity in bacterial elimination,³⁷ which could dramatically decrease the catalytic activity of engineered *E. coli* cells by killing them. To provide a more accurate explanation of this finding, we investigated the effect of indole analogues on the bacterial viability (Figure S50 in the Supporting Information). The experimental results showed that indoles substituted with Cl and Br (e.g., 5-Cl/Br and 6-Cl/Br) could completely kill *E. coli* at ~1.0 mM. The indoles substituted with F atom (e.g., 4/5/6/7-F and 5,6-difluoro) also exhibited a good bactericidal effect against *E. coli* and could completely eliminate *E. coli* at ~3.0 mM. The CH_3 - and OCH_3 -substituted indoles (e.g., 5- CH_3/OCH_3 and 6- CH_3/OCH_3) were less harmful to *E. coli*, especially the OCH_3 -substituted ones, with ~20% of *E. coli* surviving at 5.0 mM. And the effect of L-serine/PLP/sodium ascorbate/hemin/2-ketoglutaric acid on bacterial viability could be neglected (Figure S51 in the Supporting Information). Therefore, we next employed a fed-batch strategy by adding 5 mM of 6-Cl-indole in four aliquots at 2 h intervals to maintain a low indole concentration in the reaction system. As expected, the

pCDFDuet-1-Pf0A9-TDO showed an 8-fold (to ~ 3.7 mM) higher product yield than before (Figure 5C).

Considering that the potential bioactivity of KYNA derivatives may affect *E. coli* growth or production performance, we also tested the viability of *E. coli* incubated with 5.0 mM various KYNAs (Figure S52). Fortunately, no negative results were observed under this concentration, and we could confidently proceed with the subsequent experimental plan. Encouraged by these positive results, we next attempted to coexpress Pf0A9, TDO, and hKAT II in the engineered *E. coli* BL21 $\Delta tnaA$ (DE3) strain. Although hKAT II was successfully overexpressed in *E. coli*, the enzyme predominantly existed in an insoluble form. To enhance its soluble expression, we adopted a fusion strategy by fusing a highly soluble protein with its N-terminus. Both Pf0A9 and TDO are highly expressed in a soluble form in *E. coli*, especially Pf0A9, which yields approximately 70 mg per liter of TB culture after purification. Hence, Pf0A9 and TDO were fused to the N-terminus of hKAT II, respectively, via a flexible linker (Gly-Gly-Gly-Gly-Ser = GGGGS). As a result, the pCDFDuet-1-Pf0A9-GGGGS-hKAT II-TDO gave a ~ 1.5 -fold (to ~ 3.85 mM) productivity advantage over pCDFDuet-1-Pf0A9-TDO-GGGGS-hKAT II, affording the desired product 7-Cl-KYNA in 77% HPLC yield (Figures 6A and Figures S57–S60 in the Supporting Information, and Tables S12 and S13 in the Supporting Information). The pCDFDuet-1-Pf0A9-(GGGGS)₂-hKAT II-TDO with a long linker length was additionally constructed for comparison; however, the two GGGGS repeats did not lead to a significant yield improvement (Figures 6A and S61–S64 in the Supporting Information, and Tables S14 and S15 in the Supporting Information). Moreover, we also attempted to coexpress two enzymes (e.g., Pf0A9 and hKAT II, TDO, and hKAT II) from a single T7 promoter with a RBS in between (Figures 6B and S65–S68 in the Supporting Information, and Tables S16 and S17 in the Supporting Information). Unfortunately, no significant improvement in product yield was observed through this RBS engineering. We even coexpressed the Pf0A9, TDO, and hKAT II using three separate T7 promoters, however, the product yield still remained lower than that achieved with the pCDFDuet-1-Pf0A9-GGGGS-hKAT II-TDO (Figures 6B and S69 and S70 in the Supporting Information, and Table S18 in the Supporting Information). Considering the higher soluble expression level of the R20G hKAT II mutant, we then constructed the plasmid pCDFDuet-1-Pf0A9-GGGGS-R20G hKAT II-TDO (Figure 6C and Table S19 in the Supporting Information). As expected, the titer of target product 7-Cl-KYNA was further increased to ~ 4.2 mM (Figure 6C). To reduce production costs, we tested lower concentrations of the whole-cell catalysts. However, the OD₆₀₀ of 50 was already optimal (Figure 6D). In an effort to simplify the operation, we investigated different fed-batch frequencies for 6-Cl-indole addition and found that two-batch feeding gave the same yield as the four-batch control (Figure 6D). To further improve the yield of 7-Cl-KYNA, we also investigated the effect of the temperature on 7-Cl-KYNA production (Figure S71 in the Supporting Information). Through screening the reaction temperature, we found that 30 °C is the optimal temperature. The yield of 7-Cl-KYNA at 25 °C is slightly lower than that at 30 °C, while increasing the reaction temperature (e.g., 37 °C and 40 °C) leads to a more significant decrease in 7-Cl-KYNA yield. Given that endogenous amidases in *E. coli* may contribute to the final deformylation step, we next tested the

deformylation capability of *E. coli* BL21 $\Delta tnaA$ (DE3) to assess whether *E. coli* could replace TFA to accomplish this transformation (Figure S72 in the Supporting Information). Due to the instability of *N*-formyl-7-Cl-KYNA, a control experiment without *E. coli*/TFA was conducted. The experimental results showed that endogenous amidases in *E. coli* did not facilitate the deformylation of *N*-formyl-7-Cl-KYNA to 7-Cl-KYNA. However, after incubation at 30 °C for 16 h, $\sim 77\%$ spontaneous deformylation from *N*-formyl-7-Cl-KYNA to 7-Cl-KYNA was observed in the control. When the reaction time was further prolonged to 24 h, the yield of 7-Cl-KYNA increased to $\sim 84\%$. Although the *N*-formyl-7-Cl-KYNA has not been completely converted, the experimental results suggest that the amount of TFA can be further reduced (to $\sim 1/5$) in the subsequent deformylation step, as excessive TFA imposes an environmental burden, albeit at the cost of a prolonged reaction time. With the optimal reaction conditions established, the practical utility of our cascade was evaluated by performing a scale-up biotransformation experiment using whole cells coexpressing Pf0A9, TDO, and hKAT II as catalysts. Incubation of whole-cell catalysts (OD₆₀₀ = 50) with 6-Cl-indole (5 mM) and *L*-serine (5.5 mM) in 1 L scale afforded 907 mg of purified 7-Cl-KYNA (Figure 6E), demonstrating the great potential of our work in facile scale-up synthesis for future industrial applications.

4. CONCLUSION

In conclusion, we successfully established a microbial cell factory for de novo biosynthesis of KYNA derivative-based antagonist drug collection. Our approach is derived from the native *L*-Trp metabolic pathway, which was redesigned into a concise three-enzyme, one-pot cascade with extremely commercial indole analogues and *L*-serine as building blocks. The key to this streamlined process was replacing kynurenine formamidase with TFA for formyl hydrolysis and prioritizing the enzymatic deamination/cyclization step over TFA-mediated transformation, demonstrating the unprecedented flexibility in the *L*-Trp metabolic pathway and prompting a re-evaluation of its current mechanistic understanding. Appropriate enzyme mining strategies were then used to screen suitable biocatalysts for each step. All three screened enzymes (Pf0A9, TDO, and hKAT II) exhibited a relatively broader substrate spectrum (13 derivatives, including F, Cl, Br, CH₃, and OCH₃ substituents) to afford a range of substituted KYNAs in 21–85% isolated yields. Finally, integration of endogenous substrate degradation pathway deletion (*E. coli* BL21 $\Delta tnaA$ (DE3) strain), hKAT II solubility engineering (R20G hKAT II variant) and fusion-protein strategy (pCDFDuet-1-Pf0A9-GGGGS-R20G hKAT II-TDO plasmid), 907 mg of 7-Cl-KYNA was easily obtained from a 1 L-scale whole-cell reaction system, exhibiting the potential of this protocol in large-scale production.

Overall, compared to traditional chemical approaches, our *E. coli*-based biocatalytic system features milder reaction conditions and circumvent the use of large quantities of organic solvents. In addition, the excellent regio- and stereoselectivity of biocatalysis effectively prevents byproduct formation, thereby ensuring a high level of atom economy. Although the use of TFA introduces certain neutralization-related challenges, its dosage can be reduced to 0.1–0.5% or even lower due to the spontaneous deformylation that occurs at room temperature. However, as the first report on the microbial production of KYNA and its derivatives, this study

still faces considerable challenges that need to be addressed. Considering the high cost of cofactors (e.g., PLP, hemin, and α -ketoglutarate) and their increased consumption during scale-up reactions, a tailored cofactor supply system is essential to further boost the target product biosynthesis. Moreover, the high polarity of the target product hinders its efficient isolation and purification. Thus, the development of simple and customized purification strategies is also crucial for reducing the overall production costs.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acssuschemeng.6c00506>.

General procedures, experimental details, and characterization data of all reactions (e.g., ^1H NMR, ^{13}C NMR, and HRMS) (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Heng Song – College of Chemistry & Molecular Science, Wuhan University, Wuhan, Hubei 430072, China; Wuhan University Shenzhen Research Institute, Shenzhen, Guangdong 518000, China;  orcid.org/0000-0002-0049-5053; Email: hengsong@whu.edu.cn

Authors

Yunbin Wu – School of Synthetic Biology, Shanxi University, Taiyuan, Shanxi 030006, China

Li Zhao – School of Synthetic Biology, Shanxi University, Taiyuan, Shanxi 030006, China

Hao Geng – School of Synthetic Biology, Shanxi University, Taiyuan, Shanxi 030006, China

Xianquan Fu – School of Synthetic Biology, Shanxi University, Taiyuan, Shanxi 030006, China

Qinyao Jia – School of Synthetic Biology, Shanxi University, Taiyuan, Shanxi 030006, China

Wenna Gu – School of Synthetic Biology, Shanxi University, Taiyuan, Shanxi 030006, China

Zixi Shang – School of Synthetic Biology, Shanxi University, Taiyuan, Shanxi 030006, China

Complete contact information is available at:

<https://pubs.acs.org/doi/10.1021/acssuschemeng.6c00506>

Author Contributions

Y.W., L.Z., and H.G. contributed equally to this work. All authors have read, discussed, and contributed to the manuscript.

Notes

The authors declare no competing financial interest.

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